

# Rohit

B.Tech Computer Science & Applied Mathematics

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## Summary

Computer Science student with practical experience in Full-Stack Development, DevOps, and Cloud Computing. Proficient in React, Node.js, PostgreSQL, Docker, Kubernetes, Terraform, and AWS.

## Education

**B.Tech - Computer Science & Applied Mathematics**

2024 - 2028

Vivekananda Institute of Professional Studies (Affiliated to GGSIPU), New Delhi

CGPA: 7.7/10

## Technical Experience

- Developed full-stack applications using React, Node.js, Express, and PostgreSQL/MongoDB with RESTful APIs and RBAC.
- Built and containerized backend services using Docker and deployed workloads on Kubernetes using Helm.
- Designed CI/CD pipelines with Jenkins, SonarQube, Trivy, and automated deployment workflows.
- Implemented monitoring and observability using Prometheus and Grafana.
- Automated infrastructure provisioning using Terraform.

## Projects

### HealthGrid - Hospital OPD Management System

GitHub

React | Node.js | Express.js | PostgreSQL | Socket.IO | Docker

2026

- Developed a full-stack healthcare management platform with **3 role-based modules** (Receptionist, Doctor, Admin) for patient registration, consultations, and prescription management.
- Built and secured **15+ REST APIs** using Express.js, PostgreSQL, JWT authentication, and Role-Based Access Control.
- Implemented Socket.IO-based **real-time queue synchronization**, enabling instant patient status updates between reception and doctor dashboards.
- Designed an offline-first workflow using IndexedDB (Dexie.js) with PDF prescription generation, ensuring uninterrupted clinic operations during network disruptions.

### Enterprise Banking Application

GitHub

Java | Spring Boot | MySQL | Docker | Kubernetes | Maven | Terraform | Jenkins | AWS EC2

2026

- Built a **3-tier application** (UI, backend, DB) using Spring Boot and MySQL; containerized services with Docker and orchestrated using Docker Compose.
- Provisioned **dev, test, and prod environments** on AWS EC2 using Terraform, reducing infrastructure setup time from **1 hour to <10 minutes**.
- Implemented a **CI/CD pipeline** using Jenkins (build, test, deploy), enabling automated multi-environment deployments in **<5 minutes**.
- Developed Kubernetes manifests to enable container orchestration, improving scalability and deployment consistency.

### Load Balancing Simulator

GitHub

React.js | Node.js | Express.js | PostgreSQL | Docker | Kubernetes | Jenkins | Prometheus | Grafana

2026

- Built a distributed systems simulator implementing **3 load balancing algorithms** - Round Robin, Weighted Round Robin, and Least Connections - to analyze traffic flow, queue buildup, and fault tolerance.
- Developed a custom **discrete-event simulation engine** generating real-time metrics for **P95 latency, throughput, and server utilization** with interactive dashboards.
- Automated CI/CD using **Jenkins, SonarQube, and Trivy**, reducing deployment and security validation time from **35 minutes to <5 minutes**.
- Containerized services with Docker and deployed on Kubernetes using **Prometheus + Grafana monitoring**, reducing environment setup time from **~25 minutes to <4 minutes**.

## Technical Skills

**Languages:** JavaScript, Java, Bash, YAML

**Frontend:** HTML, CSS, React, Tailwind CSS, Vite

**Backend:** Node.js, Express.js

**Databases:** PostgreSQL, MySQL, MongoDB

**Cloud & DevOps:** AWS (EC2, S3, DynamoDB), Docker, Kubernetes, Helm, Jenkins

**IaC & Automation:** Terraform, Ansible, Git

**Monitoring & Security:** Prometheus, Grafana, SonarQube, Trivy, OWASP, JWT, OAuth, RBAC

**Tools:** Linux, Postman, Docker Hub, VS Code, Vim